

3-6-6

1. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant. Find $f'(x)$ and $f''(x)$.

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} = -\frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $f'(x)$

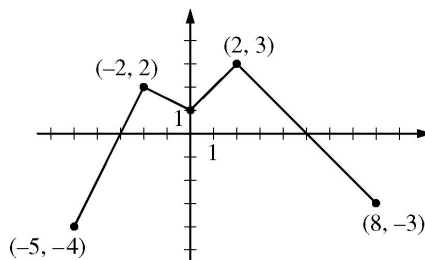
1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} = -\frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

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2.

Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) dt$.

Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.

Part B

The response can earn up to 3 points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$$g''(-2) = f'(-2) \text{ does not exist.}$$



0	1	2	3
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The response earns all three of the following points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$$g''(-2) = f'(-2) \text{ does not exist.}$$

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3. If $\frac{dy}{dx} = \frac{-3}{\sqrt{x}}$ for $x > 0$, then $\frac{d^3y}{dx^3} =$

(A) $\frac{-27}{\sqrt{x^3}}$

(B) $\frac{-9}{4\sqrt{x^5}}$

(C) $\frac{3}{4\sqrt{x^3}}$

(D) $\frac{45}{8\sqrt{x^7}}$

Answer B

Correct. Since the first derivative is given and the third derivative is to be found, $\frac{dy}{dx}$ must be differentiated twice using the power rule.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (-3x^{-1/2}) = \frac{3}{2}x^{-3/2}$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d}{dx} \left(\frac{3}{2}x^{-3/2} \right) = -\frac{9}{4}x^{-5/2} = -\frac{9}{4\sqrt{x^5}}$$

4. Let f be the function given by the $f(x) = \frac{\ln x}{x}$ for all $x > 0$. The derivative of f is given by $f'(x) = \frac{1-\ln x}{x^2}$.

The graph of the function f has exactly one point of inflection. Find the x -coordinate of this point.

Part C

The response can earn up to 3 points:

2 point: For $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2\ln x}{x^3} \text{ for all } x > 0 \quad f''(x) = 0 \text{ when } -3 + 2\ln x = 0 \Rightarrow x = e^{3/2}$$

1 point: For the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$



0	1	2	3
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The response earns all three of the following points:

2 point: For $f''(x)$

$$f''(x) = \frac{-\frac{1}{x}x^2 - (1 - \ln x)2x}{x^4} = \frac{-3 + 2\ln x}{x^3} \text{ for all } x > 0 \quad f''(x) = 0 \text{ when } -3 + 2\ln x = 0 \Rightarrow x = e^{3/2}$$

1 point: For the correct answer

The graph of f has a point of inflection at $x = e^{3/2}$ because $f''(x)$ changes sign at $x = e^{3/2}$

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5. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
For what values of x is the graph of f concave down?

Part C

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

$$\begin{aligned} 2e^{-x}(x-2) &< 0 \\ (x-2) &< 0 \\ x &< 2 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\ &= 2e^{-x}(x-2) \end{aligned}$$

Concave down when

$$\begin{aligned} 2e^{-x}(x-2) &< 0 \\ (x-2) &< 0 \\ x &< 2 \end{aligned}$$

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6. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	1	2
$g(x)$	6	4	2	-1
$g'(x)$	-1	-7	-2	-3

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(1) = 0$ and derivative given by $f'(x) = x \sin(x^2)$.

- Find $f''(e^{1.5})$. Express your answer as a decimal approximation.
- Let h be the function defined by $h(x) = f(g(3x))$. Find $h'(0)$. Express your answer as a decimal approximation.
- Write an equation for the line tangent to the graph of g^{-1} , the inverse function of g , at $x = -1$.
- The point $(1, 4)$ lies on the curve in the xy -plane given by the equation $f(x) + (g(x))^2 = xy$. What is the value of $\frac{dy}{dx}$ at the point $(1, 4)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes a correct decimal approximation of $f''(e^{1.5})$

Solution:

$$f''(e^{1.5}) = 14.144$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = f'(g(3x)) \cdot g'(3x) \cdot 3$$

$$\begin{aligned} h'(0) &= f'(g(0)) \cdot g'(0) \cdot 3 = f'(4) \cdot (-7) \cdot 3 \\ &= 4 \sin 16 \cdot (-21) = -84 \sin 16 \\ &= 24.184 \text{ (or } 24.183) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

Because $g(2) = -1$, $g^{-1}(-1) = 2$.

$$(g^{-1})'(-1) = \frac{1}{g'(g^{-1}(-1))} = \frac{1}{g'(2)} = \frac{1}{-3}$$

An equation for the tangent line is $y = 2 - \frac{1}{3}(x + 1)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error.

Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}(f(x) + (g(x))^2) = \frac{d}{dx}(xy)$$

$$\Rightarrow f'(x) + 2g(x) \cdot g'(x) = y + x \frac{dy}{dx}$$

$$\Rightarrow f'(1) + 2g(1) \cdot g'(1) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \sin 1 + 2 \cdot 2 \cdot (-2) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \sin 1 - 12 = -11.159 \text{ (or } -11.158)$$

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7. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	6
$f(x)$	-5	4	1	7
$f'(x)$	-1	2	-2	4

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(x^2 - x)$.

- What is the value of $g'(3)$?
- It is known that $g''(0) = -1$. What is the value of $f''(0)$?
- Is there a value c , for $0 < c < 3$, such that $g(c) = 5$? Justify your answer.
- Let h be the function with derivative given by $h'(x) = 4e^{\cos x}$. At what value of x in the interval $-3 \leq x \leq 0$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $-3 \leq x \leq 0$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes one of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(3)$

Solution:

$$g'(x) = f'(x^2 - x)(2x - 1)$$

$$g'(3) = f'(6)(5) = 4(5) = 20$$

Part B

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Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(x^2 - x)(2x - 1)^2 + f'(x^2 - x)(2)$$

$$g''(0) = f''(0)(-1)^2 + f'(0)(2)$$

$$-1 = f''(0) + 2(2) \Rightarrow f''(0) = -5$$

Part C

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
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The student response accurately includes both of the criteria below.

- ☐ g is continuous, with reason
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = 4 \text{ and}$$

$$g(3) = f(6) = 7.$$

$$\Rightarrow g(0) < 5 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 5$.

Part D

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At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

The average rate of change of f over the interval $-3 \leq x \leq 0$

$$\frac{f(0)-f(-3)}{0-(-3)} = \frac{4-(-5)}{3} = 3.$$

$$h'(x) = 4e^{\cos x} = 3 \text{ for } -3 \leq x \leq 0. \Rightarrow x = -1.862602$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $-3 \leq x \leq 0$ at $x = -1.863$ (or -1.862).

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8. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

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Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	2	3
$g(x)$	4	1	0	-2
$g'(x)$	-2	-6	-3	-4

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(3) = 2$ and derivative given by $f'(x) = x \cos(e^x)$.

- Find $f''(\sin 2)$. Express your answer as a decimal approximation.
- Let h be the function defined by $h(x) = g(f(2x))$. Find $h'(1.5)$. Express your answer as a decimal approximation.
- Write an equation for the line tangent to the graph of g^{-1} , the inverse function of g , at $x = -2$.
- The point $(3, 1)$ lies on the curve in the xy -plane given by the equation $(f(x))^2 + g(x) = xy - 1$. What is the value of $\frac{dy}{dx}$ at the point $(3, 1)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
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The student response accurately includes a correct decimal approximation of $f''(\sin 2)$

Solution:

$$f''(\sin 2) = -2.173 \text{ (or } -2.172)$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = g'(f(2x)) \cdot f'(2x) \cdot 2$$

$$\begin{aligned} h'(1.5) &= g'(f(3)) \cdot f'(3) \cdot 2 = g'(2) \cdot (3 \cos(e^3)) \cdot 2 \\ &= -3(3 \cos(e^3)) \cdot 2 \\ &= -18 \cos(e^3) = -5.915 \text{ (or } -5.914) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\text{Because } g(3) = -2, g^{-1}(-2) = 3.$$

$$(g^{-1})'(-2) = \frac{1}{g'(g^{-1}(-2))} = \frac{1}{g'(3)} = -\frac{1}{4}$$

$$\text{An equation for the tangent line is } y = 3 - \frac{1}{4}(x + 2).$$

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error. Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

implicit differentiation (2 points for full credit)

3-6-6☐☐

answer

Solution:

$$\frac{d}{dx} \left((f(x))^2 + g(x) \right) = \frac{d}{dx} (xy - 1)$$

$$\Rightarrow 2f(x) \cdot f'(x) + g'(x) = y + x \frac{dy}{dx}$$

$$\Rightarrow 2f(3) \cdot f'(3) + g'(3) = 1 + 3 \frac{dy}{dx}$$

$$\Rightarrow 2(2) \cdot (3 \cos(e^3)) + (-4) = 1 + 3 \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -0.352$$

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9.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

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Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	0	3	4
$f(x)$	5	-1	2	7
$f'(x)$	-2	4	0	1

The table above gives values of a twice-differentiable function f and its first derivative f' for selected values of x . Let g be the function defined by $g(x) = f(2x - x^2)$.

- What is the value of $g'(-1)$?
- It is known that $g''(0) = 0$. What is the value of $f''(0)$?
- Is there a value c , for $0 < c < 3$, such that $g(c) = 2$? Justify your answer.
- Let h be the function with first derivative given by $h'(x) = 4xe^x$. At what value of x in the interval $0 \leq x \leq 4$ does the instantaneous rate of change of h equal the average rate of change of f over the interval $0 \leq x \leq 4$?

Part A

Neither point may be earned for a response that does not apply the chain rule. The second point may be earned with a maximum of one error in $g'(x)$, unless the error is failing to apply the chain rule. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $g'(x)$
- ☐ $g'(-1)$

Solution:

$$g'(x) = f'(2x - x^2)(2 - 2x)$$

$$g'(-1) = f'(-3)(4) = (-2)(4) = -8$$

Part B

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Points may be earned in this part for consistent and correct use of the $g'(x)$ from part (a).

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $g''(x)$ (2 points for full credit)
- ☐ answer

Solution:

$$g''(x) = f''(2x - x^2)(2 - 2x)^2 + f'(2x - x^2)(-2)$$

$$g''(0) = f''(0)(2)^2 + f'(0)(-2)$$

$$0 = 4f''(0) + 4(-2) \Rightarrow f''(0) = 2$$

Part C

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
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The student response accurately includes both of the criteria below.

- ☐ g is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

f is twice differentiable. $\Rightarrow f$ is continuous.

g is continuous as the composition of continuous functions.

$$g(0) = f(0) = -1 \text{ and } g(3) = f(-3) = 5.$$

$$\Rightarrow g(0) < 2 < g(3)$$

By the Intermediate Value Theorem, there is a value of c , for $0 < c < 3$, such that $g(c) = 2$.

Part D

At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.

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At most 1 out of 2 points is earned for an answer based on an average rate of change with a maximum of one computational error. Substitution of function values is required.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ average rate of change
- ☐ answer

Solution:

The average rate of change of f over the interval $0 \leq x \leq 4$ is $\frac{f(4)-f(0)}{4-0} = \frac{7-(-1)}{4} = 2$.

$$h'(x) = 4xe^x = 2 \text{ for } 0 \leq x \leq 4. \Rightarrow x = 0.351734$$

The instantaneous rate of change of h equals the average rate of change of f over the interval $0 \leq x \leq 4$ at $x = 0.352$ (or 0.351).

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10. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 3$. The derivative of f is given by $f'(x) = \cos(\pi x) + x^4 + 6$.

- (a) Find $f''(-3)$.
- (b) Write an equation for the line tangent to the graph of $y = (f(x))^2$ at $x = 0$.
- (c) Let g be the function defined by $g(x) = f(\sqrt{2x^2 + 7})$. Find $g'(3)$.
- (d) Let h be the inverse function of f . Find $h'(3)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = -\pi \sin(\pi x) + 4x^3$$

$$f''(-3) = -\pi \sin(-3\pi) + 4(-3)^3 = -108$$

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error. Trigonometric function values do not need to be evaluated. The third point may be earned if response contains a slope that is consistent with the derivative expression. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response does not accurately include any of the criteria below.

☐ $\frac{d}{dx}((f(x))^2)$

3-6-6

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\text{At } x = 0, y = (f(0))^2 = 3^2 = 9.$$

$$\frac{d}{dx}((f(x))^2) = 2f(x)f'(x)$$

$$\left. \frac{d}{dx}((f(x))^2) \right|_{x=0} = 2f(0)f'(0) = 2(3)(\cos 0 + 6) = 6(\cos 0 + 6) = 42$$

An equation for the tangent line is $y = 9 + 42x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. Trigonometric function values do not need to be evaluated. A simplified answer is not required. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'(x) = f'(\sqrt{2x^2 + 7}) \cdot \frac{4x}{2\sqrt{2x^2 + 7}} = f'(\sqrt{2x^2 + 7}) \cdot \frac{2x}{\sqrt{2x^2 + 7}}$$

$$g'(3) = f'(\sqrt{2 \cdot 3^2 + 7}) \cdot \frac{2 \cdot 3}{\sqrt{2 \cdot 3^2 + 7}} = f'(5) \cdot \frac{6}{5} = (\cos(5\pi) + 631) \cdot \frac{6}{5} = (630)\frac{6}{5} = 756$$

Part D

Substitution of function values is required. Trigonometric function values do not need to be evaluated. Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ answer

Solution:

$$\text{Because } f(0) = 3, h(3) = 0.$$

3-6-6

$$h'(3) = \frac{1}{f'(h(3))} = \frac{1}{f'(0)} = \frac{1}{(\cos 0 + 6)} = \frac{1}{7}$$

3-6-6

11. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let f be an increasing function with $f(0) = 2$. The derivative of f is given by $f'(x) = \sin(\pi x) + x^2 + 3$.

- Find $f''(-2)$.
- Write an equation for the line tangent to the graph of $y = \frac{1}{f(x)}$ at $x = 0$.
- Let g be the function defined by $g(x) = f(\sqrt{3x^2 + 4})$. Find $g'(2)$.
- Let h be the inverse function of f . Find $h'(2)$.

Part A

The second point cannot be earned without the first point. The second point does not require a simplified answer; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ chain rule
- ☐ answer

Solution:

$$f''(x) = \pi \cos(\pi x) + 2x$$

$$f''(-2) = \pi \cos(-2\pi) + 2(-2) = \pi - 4$$

Part B

The second point may be earned based on a derivative expression that includes application of power rule and chain rule with a maximum of one computational error.

The third point may be earned if response contains a slope that is consistent with the derivative expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3
---	---	---	---

3-6-6

The student response accurately includes all three of the criteria below.

- ☐ $\frac{d}{dx}\left(\frac{1}{f(x)}\right)$
- ☐ slope
- ☐ equation for tangent line

Solution:

$$\text{At } x = 0, y = \frac{1}{f(0)} = \frac{1}{2}.$$

$$\frac{d}{dx}\left(\frac{1}{f(x)}\right) = \frac{d}{dx}(f(x))^{-1} = -(f(x))^{-2}f'(x) = -\frac{f'(x)}{(f(x))^2}$$

$$\left.\frac{d}{dx}\left(\frac{1}{f(x)}\right)\right|_{x=0} = -\frac{f'(0)}{(f(0))^2} = -\frac{3}{4}$$

An equation for the tangent line is $y = \frac{1}{2} - \frac{3}{4}x$.

Part C

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$g'(x) = f'(\sqrt{3x^2 + 4}) \cdot \frac{6x}{2\sqrt{3x^2 + 4}} = f'(\sqrt{3x^2 + 4}) \cdot \frac{3x}{\sqrt{3x^2 + 4}}$$

$$g'(2) = f'(\sqrt{3 \cdot 2^2 + 4}) \cdot \frac{3 \cdot 2}{\sqrt{3 \cdot 2^2 + 4}} = f'(4) \cdot \frac{3}{2} = 19 \cdot \frac{3}{2} = \frac{57}{2}$$

Part D

Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1
---	---



The student response accurately includes a correct value of $h'(2)$

3-6-6

Solution:

Because $f'(0) = 2$, $h'(2) = 0$.

$$h'(2) = \frac{1}{f'(h(2))} = \frac{1}{f'(0)} = \frac{1}{3}$$

12. Consider the differential equation $\frac{dy}{dx} = 2x - y$

Find d^2y/dx^2 in terms of x and y . Determine the concavity of all solution curves for the given differential equation in Quadrant II. Give a reason for your answer.

Part B

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$.

Therefore, all solution curves are concave up in Quadrant II.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

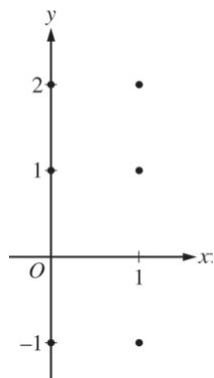
In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$.

Therefore, all solution curves are concave up in Quadrant II.

3-6-6

13. Consider the differential equation $\frac{dy}{dx} = 2x - y$.

(a) On the axes provided, sketch a slope field for the given differential equation at the six points indicated.

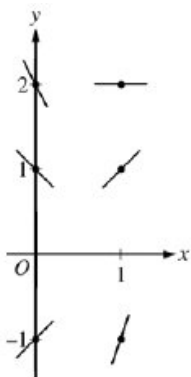


- (b) Find $\frac{d^2y}{dx^2}$ in terms of x and y . Determine the concavity of all solution curves for the given differential equation in Quadrant II. Give a reason for your answer.
- (c) Let $y = f(x)$ be the particular solution to the differential equation with the initial condition $f(2) = 3$. Does f have a relative minimum, a relative maximum, or neither at $x = 2$? Justify your answer.
- (d) Find the values of the constants m and b for which $y = mx + b$ is a solution to the differential equation.

Part A

1 point is earned for slopes where $x = 0$

1 point is earned for slopes where $x = 1$



✓

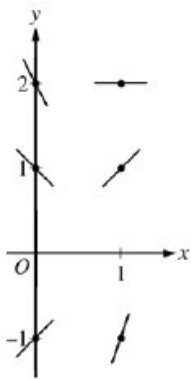
0	1	2
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The student response earns all of the following points:

1 point is earned for slopes where $x = 0$

1 point is earned for slopes where $x = 1$

3-6-6



Part B

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$. Therefore, all solution curves are concave up in Quadrant II.



0	1	2
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The student response earns all of the following points:

1 point is earned for $\frac{d^2y}{dx^2}$

1 point is earned for concave up with reason

$$\frac{d^2y}{dx^2} = 2 - \frac{dy}{dx} = 2 - (2x - y) = 2 - 2x + y$$

In Quadrant II, $x < 0$ and $y > 0$, so $2 - 2x + y > 0$. Therefore, all solution curves are concave up in Quadrant II.

Part C

1 point is earned for considering $\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\left. \frac{dy}{dx} \right|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore, f has neither a relative minimum nor a relative maximum at $x = 2$.



0	1	2
---	---	---

3-6-6

The student response earns all of the following points:

1 point is earned for considering $\frac{dy}{dx}\big|_{(x,y)=(2,3)}$

1 point is earned for the conclusion with justification

$$\frac{dy}{dx}\big|_{(x,y)=(2,3)} = 2(2) - 3 = 1 \neq 0$$

Therefore, f has neither a relative minimum nor a relative maximum at $x = 2$.

Part D

1 point is earned for $\frac{d}{dx}(mx + b) = m$

1 point is earned for $2x - y = m$

1 point is earned for answer

$$y = mx + b \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(mx + b) = m$$

$$2x - y = m$$

$$2x - (mx + b) = m$$

$$(2 - m)x - (m + b) = 0$$

$$2 - m = 0 \Rightarrow m = 2$$

$$b = -m \Rightarrow b = -2$$

Therefore, $m = 2$, and $b = -2$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}(mx + b) = m$

1 point is earned for $2x - y = m$

1 point is earned for answer

$$y = mx + b \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(mx + b) = m$$

$$2x - y = m$$

$$2x - (mx + b) = m$$

$$(2 - m)x - (m + b) = 0$$

$$2 - m = 0 \Rightarrow m = 2$$

$$b = -m \Rightarrow b = -2$$

Therefore, $m = 2$, and $b = -2$

3-6-6

14. Consider the logistic differential equation $\frac{dy}{dt} = \frac{y}{8}(6 - y)$. Let $y = f(t)$ be the particular solution to the differential equation with $f(0) = 8$.

Write the second-degree Taylor polynomial for f about $t = 0$, and use it to approximate $f(1)$.

Part C

The response can earn up to 4 points:

2 points: $\frac{d^2y}{dt^2}$

1 point: second-degree Taylor polynomial

1 point: approximation of $f(1)$

$$\frac{d^2y}{dt^2} = \frac{1}{8} \frac{dy}{dt} (6 - y) + \frac{y}{8} \left(-\frac{dy}{dt} \right) f(0) = 8; \quad f'(0) = \frac{dy}{dt} \Big|_{t=0} = \frac{8}{8}(6 - 8) = -2; \quad \text{and} \quad f''(0) = \frac{d^2y}{dt^2} \Big|_{t=0} = \frac{1}{8}(-2)(-2) + \frac{8}{8}(2) = \frac{5}{2}$$

The second-degree Taylor polynomial for f about

$$t = 0 \text{ is } P_2(t) = 8 - 2t + \frac{5}{4}t^2. \quad f(1) \approx P_2(1) = \frac{29}{4}$$



0	1	2	3	4
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The response can earn up to 4 points:

2 points: $\frac{d^2y}{dt^2}$

1 point: second-degree Taylor polynomial

1 point: approximation of $f(1)$

$$\frac{d^2y}{dt^2} = \frac{1}{8} \frac{dy}{dt} (6 - y) + \frac{y}{8} \left(-\frac{dy}{dt} \right) f(0) = 8; \quad f'(0) = \frac{dy}{dt} \Big|_{t=0} = \frac{8}{8}(6 - 8) = -2; \quad \text{and} \quad f''(0) = \frac{d^2y}{dt^2} \Big|_{t=0} = \frac{1}{8}(-2)(-2) + \frac{8}{8}(2) = \frac{5}{2}$$

The second-degree Taylor polynomial for f about

$$t = 0 \text{ is } P_2(t) = 8 - 2t + \frac{5}{4}t^2. \quad f(1) \approx P_2(1) = \frac{29}{4}$$

3-6-6

15. If $x^2 + y^2 = 25$, what is the value of $\frac{d^2y}{dx^2}$ at the point (4,3) ?

(A) $-\frac{25}{27}$

(B) $-\frac{7}{27}$

(C) $\frac{7}{27}$

(D) $\frac{3}{4}$

(E) $\frac{25}{27}$



16. If $y = 2 \cos\left(\frac{x}{2}\right)$, then $\frac{d^2y}{dx^2} =$

(A) $-8 \cos\left(\frac{x}{2}\right)$

(B) $-2 \cos\left(\frac{x}{2}\right)$

(C) $-\sin\left(\frac{x}{2}\right)$

(D) $-\cos\left(\frac{x}{2}\right)$

(E) $-\frac{1}{2} \cos\left(\frac{x}{2}\right)$



17. If $y = e^{nx}$, then $\frac{d^ny}{dx^n} =$

(A) $n^n e^{nx}$

(B) $n! e^{nx}$

(C) ne^{nx}

(D) $n^n e^x$

(E) $n! e^x$



18. If $y = \sin x$ and $y^{(n)}$ means the n th derivative of y with respect to x then the smallest positive integer n for which $y^{(n)} = y$ is

(A) 2

(B) 4

(C) 5

(D) 6

(E) 8



19. If $f(x) = (2x + 1)^4$, then the 4th derivative of $f(x)$ at $x = 0$ is

(A) 0

(B) 24

(C) 48

(D) 240

(E) 384



3-6-6

20. If $y = \cos x - \ln(2x)$, then $\frac{d^3y}{dx^3} =$

(A) $\sin x - \frac{2}{x^3}$

(B) $-\sin x - \frac{2}{x^3}$

(C) $\sin x - \frac{1}{x^3}$

(D) $-\sin x - \frac{1}{x^3}$

Answer A

This option is correct. This question requires repeated differentiation using knowledge of the derivatives of sine and cosine, the derivative of the natural logarithm, and the power rule for derivatives, as well as recognizing the need to use the chain rule for the first derivative.

$$\frac{dy}{dx} = -\sin x - \frac{2}{2x} = -\sin x - \frac{1}{x}$$

$$\frac{d^2y}{dx^2} = -\cos x + \frac{1}{x^2}$$

$$\frac{d^3y}{dx^3} = \sin x - \frac{2}{x^3}$$

21. If $\frac{dy}{dx} = x^4 - 2x^3 + 3x - 1$, then $\frac{d^3y}{dx^3}$ evaluated at $x=2$ is

(A) 11

(B) 24

(C) 26

(D) 125

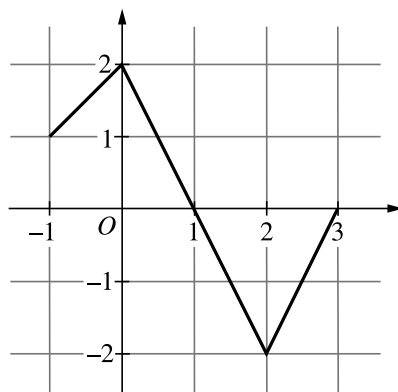
3-6-6

22. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on the closed interval $[-1, 3]$ whose graph, consisting of three line segments, is shown.

Let g be the function given by $g(x) = \int_0^x f(t) \, dt$.

- On what intervals, if any, is g decreasing? Give a reason for your answer.
- Find $g''(-\frac{1}{2})$, or explain why it does not exist.
- What is the absolute minimum value of g on the closed interval $[-1, 3]$? Justify your answer.
- Let h be the function defined by $h(x) = \ln(g(x))$ for $0 < x < 2$. Find $h'(\frac{3}{2})$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f(x)$ in any part of the problem.

A response of $g''(x) = f'(x)$ or $g''(-\frac{1}{2}) = f'(-\frac{1}{2})$ in part (b) is sufficient to earn the first point.

The interval $[1, 3]$ can be reported $(1, 3)$, $(1, 3]$, or $[1, 3]$.

The second point can be earned without the first by presenting a correct interval and a statement that $f(x) < 0$ on this interval.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $g'(x) = f(x)$ in any part of response
- ☐ Answer with reason

3-6-6

Solution:

$g'(x) = f(x) < 0$ on the open interval $(1, 3)$. Therefore, g is decreasing on the closed interval $[1, 3]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 1 with no supporting work earns the point.

Any work shown must be correct in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''\left(-\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right) = 1$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for any presented consideration of $x = -1$ and $x = 3$.

A response that finds only that $g(-1) = -1.5$ and declares this as the minimum earns only the second point.

A response that evaluates $g(x)$ at any values of x in addition to $x = -1$, $x = 1$, and $x = 3$ does not earn the third point.

To earn the third point, a response must:

- have earned the first point,
- have no errors in the evaluation of $g(-1)$ and $g(3)$,
- conclude that $g(3) > g(-1)$, and
- either evaluate $g(1)$ as part of the Candidates Test or include a global argument that specifically addresses the behavior of $g(x)$ at $x = 1$ (e.g., “ $g'(x) = f(x)$ changes signs from positive to negative at $x = 1$ ” or “ $g'(1) = f(1) = 0$ and there is a maximum at $x = 1$ ”).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -1$ and $x = 3$
- ☐ Answer
- ☐ Justification

3-6-6

Solution:

Because $g'(x) = f(x) > 0$ for $-1 < x < 1$ and $g'(x) = f(x) < 0$ for $1 < x < 3$, the minimum value of g must occur at either $x = -1$ or $x = 3$.

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Alternate solution using Candidates Test:

$$g'(x) = f(x) = 0 \Rightarrow x = 1$$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(1) = \int_0^1 f(t) dt = 1$$

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = \frac{1}{g(x)} \cdot g'(x)$, $h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right)$, or $\frac{1}{\frac{3}{4}} \cdot (-1)$.

To earn the second point a response must use the correct value of either $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$ (or both) in an expression for $h'\left(\frac{3}{2}\right)$.

The minimal answer that earns all 3 points is $\frac{1}{\frac{3}{4}} \cdot (-1)$.

An answer of $-\frac{4}{3}$ with incorrect or without supporting work does not earn any points.

The third point is earned only for $-\frac{4}{3}$ or equivalent.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $h'(x)$
- ☐ Correct value of $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$
- ☐ Answer

Solution:

3-6-6

$$h'(x) = \frac{1}{g(x)} \cdot g'(x)$$

$$h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right) = \frac{1}{\frac{3}{4}} \cdot (-1) = -\frac{4}{3}$$

23. If $f(x) = e^{7x}$, then $f^{(7)}(x) =$

(A) $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$

(B) $\frac{1}{7^7}e^{7x}$

(C) e^{7x}

(D) 7^7e^{7x}

**Answer D**

Correct. The notation $f^{(7)}(x)$ denotes the seventh derivative of f . The first three derivatives are $f'(x) = 7e^{7x}$, $f''(x) = 7 \cdot (7e^{7x})$, and $f'''(x) = 7 \cdot (7 \cdot 7e^{7x})$. Following this pattern, the chain rule must be applied seven times while computing $f^{(7)}(x)$, and so $f^{(7)}(x) = 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot 7 \cdot e^{7x} = 7^7e^{7x}$.

24. If $y = e^{2 \sin x}$, then $\frac{d^2y}{dx^2} =$

(A) $(4\cos^2x)(e^{4 \sin x})$

(B) $(-4 \sin x \cos x)(e^{2 \sin x})$

(C) $(-\sin x + \cos x)(2e^{2 \sin x})$

(D) $(-\sin x + 2\cos^2x)(2e^{2 \sin x})$

**Answer D**

Correct. The notation $\frac{d^2y}{dx^2}$ indicates the second derivative of y . Computing the first and second derivatives requires the use of both the chain rule and the product rule.

$$\frac{dy}{dx} = 2 \cos x \cdot e^{2 \sin x}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (2 \cos x \cdot e^{2 \sin x}) = -2 \sin x \cdot e^{2 \sin x} + 4 \cos^2 x \cdot e^{2 \sin x} = (-\sin x + 2 \cos^2 x)(2e^{2 \sin x})$$

3-6-6

25. Let f be the function given by $f(x) = \sin x + e^{-x} + 3x$. Which of the following statements is true for $y = f(x)$?

(A) $y'' = \sin x + e^{-x}$

(B) $\frac{d^3y}{dx^3} = \frac{dy}{dx}$

(C) $f^{(4)}(x) = f'(x) \cdot f'''(x)$

(D) $y - \frac{d^4y}{dx^4} = 3x$

**Answer D**

Correct. The notation $\frac{d^4y}{dx^4}$ indicates the fourth derivative of y .

$$\frac{dy}{dx} = \cos x - e^{-x} + 3$$

$$\frac{d^2y}{dx^2} = -\sin x + e^{-x}$$

$$\frac{d^3y}{dx^3} = -\cos x - e^{-x}$$

$$\frac{d^4y}{dx^4} = \sin x + e^{-x}$$

$$\text{Then } y - \frac{d^4y}{dx^4} = (\sin x + e^{-x} + 3x) - (\sin x + e^{-x}) = 3x.$$

26. If $y = 3e^{-2x}$, then $\frac{d^3y}{dx^3} =$

(A) $-24e^{-2x}$

(B) $-6e^{-2x}$

(C) $48e^{-2x}$

(D) $-216e^{-6x}$

**Answer A**

Correct. This notation represents the third derivative of y . Repeated differentiation produces the following.

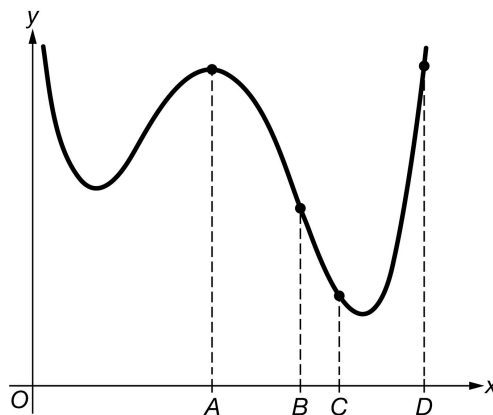
$$\frac{dy}{dx} = -6e^{-2x}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} (-6e^{-2x}) = 12e^{-2x}$$

$$\frac{d^3y}{dx^3} = \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d}{dx} (12e^{-2x}) = -24e^{-2x}$$

3-6-6

27.

Graph of f'

The figure above shows the graph of f' , the derivative of the function f . At which of the four indicated values of x is $f''(x)$ least?

- (A) A
 (B) B
 (C) C
 (D) D

**Answer B**

Correct. The second derivative f'' is the rate of change of the first derivative f' . The value of $f''(x)$ therefore represents the slope of the line tangent to the graph of f' at x . The tangent line at $x = A$ is horizontal, so $f''(A) = 0$. The tangent lines at $x = B$ and $x = C$ both have negative slopes, but the tangent line at $x = B$ is steeper than the tangent line at $x = C$. Therefore, $f''(B)$ is less than $f''(C)$. The tangent line at $x = D$ has a positive slope, so $f''(D) > 0$. Therefore $f''(B)$ has the least value.

28. Let $y = f(x)$ be a twice-differentiable function such that $f(-1) = 5$ and $\frac{dy}{dx} = \frac{1}{5}(xy^2 + 4y)^2$. What is the value of $\frac{d^2y}{dx^2}$ at $x = -1$?
- (A) -190
 (B) -70
 (C) -2
 (D) 10

**Answer D**

Correct. The second derivative can be found by using implicit differentiation of the first derivative and then evaluating at the point $(-1, 5)$.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{1}{5}(xy^2 + 4y)^2 \right) = \frac{2}{5}(xy^2 + 4y) \left(y^2 + 2xy \frac{dy}{dx} + 4 \frac{dy}{dx} \right)$$

$$\text{Since } \frac{dy}{dx} \Big|_{(-1,5)} = \frac{1}{5}(-25 + 20)^2 = 5,$$

$$\frac{d^2y}{dx^2} \Big|_{(-1,5)} = \frac{2}{5}(xy^2 + 4y) \left(y^2 + 2xy \frac{dy}{dx} + 4 \frac{dy}{dx} \right) \Big|_{(-1,5)} = \frac{2}{5}(-5)(25 + (-10)(5) + 4(5)) = 10.$$

3-6-6

29. Let f be the function given by $f(x) = -\frac{32}{x} + x^3$ on the closed interval $[1, 4]$. What is the instantaneous rate of change of f' at $x = 2$?
- (A) -8
- (B) 4 ✓
- (C) 5
- (D) 20

Answer B

Correct. The instantaneous rate of change of the first derivative is the second derivative.

$$f'(x) = \frac{32}{x^2} + 3x^2$$

$$f''(x) = \frac{d}{dx}(f'(x)) = \frac{-64}{x^3} + 6x$$

$$f''(2) = -8 + 12 = 4$$

30. Let g be the function given by $g(x) = \sin(-x) + \cos x - 10$. Which of the following statements is true for $y = g(x)$?

(A) $y + 10 = \frac{d^4 y}{dx^4}$ ✓

(B) $g^{(4)}(x) = (g''(x))^2$

(C) $g''(x) - 10 = g(x)$

(D) $y'' = \sin(-x) + \cos x$

Answer A

Correct. The notation $\frac{d^4 y}{dx^4}$ indicates the fourth derivative of y .

$$\frac{dy}{dx} = -\cos(-x) - \sin x$$

$$\frac{d^2 y}{dx^2} = -\sin(-x) - \cos x$$

$$\frac{d^3 y}{dx^3} = \cos(-x) + \sin x$$

$$\frac{d^4 y}{dx^4} = \sin(-x) + \cos x$$

$$\text{Then } y + 10 = (\sin(-x) + \cos x - 10) + 10 = \sin(-x) + \cos x = \frac{d^4 y}{dx^4}.$$

3-6-6

31. If $y = -3 \cos(2x)$, then $\frac{d^2 y}{dx^2} =$

(A) $-12 \cos(2x)$

(B) $12 \cos(2x)$ ✓

(C) $-3 \cos(2x)$

(D) $3 \cos(2x)$

Answer B

Correct. The chain rule must be used twice to find the second derivative.

$$\frac{dy}{dx} = -3(-\sin(2x)) \cdot 2 = 6 \sin(2x)$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx}(6 \sin(2x)) = 6 \cos(2x) \cdot 2 = 12 \cos(2x)$$

32. If $y = 2 \ln x$, then $\frac{d^4 y}{dx^4} =$

(A) $\frac{2}{x}$

(B) $-\frac{12}{x^4}$ ✓

(C) $\frac{16}{x^4}$

(D) $\frac{48}{x^5}$

Answer B

Correct. This notation represents the fourth derivative of y . Repeated differentiation produces the following.

$$\frac{dy}{dx} = \frac{2}{x}$$

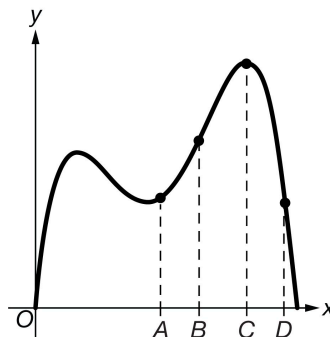
$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{2}{x} \right) = -\frac{2}{x^2}$$

$$\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right) = \frac{d}{dx} \left(-\frac{2}{x^2} \right) = \frac{4}{x^3}$$

$$\frac{d^4 y}{dx^4} = \frac{d}{dx} \left(\frac{d^3 y}{dx^3} \right) = \frac{d}{dx} \left(\frac{4}{x^3} \right) = -\frac{12}{x^4}$$

3-6-6

33.

Graph of f'

The figure above shows the graph of f' , the derivative of the function f . At which of the four indicated values of x is $f''(x)$ greatest?

- (A) A
 (B) B
 (C) C
 (D) D

**Answer B**

Correct. The second derivative f'' is the rate of change of the first derivative f' . The value of $f''(x)$ therefore represents the slope of the line tangent to the graph of f' at x . The tangent lines at $x = A$ and $x = B$ both have positive slopes, but the tangent line at $x = B$ is steeper than the tangent line at $x = A$. Therefore $f''(B)$ is greater than $f''(A)$. The tangent line at $x = C$ is horizontal, so $f''(C) = 0$. The tangent line at $x = D$ has a negative slope, so $f''(D) < 0$. Therefore $f''(B)$ has the greatest value.

34. Let $y = f(x)$ be a twice-differentiable function such that $f(1) = 3$ and $\frac{dy}{dx} = 4\sqrt{y^2 + 7x^2}$. What is the value of $\frac{d^2y}{dx^2}$ at $x = 1$?
- (A) 10
 (B) 23
 (C) 55
 (D) 160

**Answer C**

Correct. The second derivative can be found using implicit differentiation of the first derivative, then evaluated at the point $(1, 3)$.

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(4\sqrt{y^2 + 7x^2} \right) = 2 \cdot \frac{2y \frac{dy}{dx} + 14x}{\sqrt{y^2 + 7x^2}}$$

Since $\frac{dy}{dx} \Big|_{(1,3)} = 4\sqrt{3^2 + 7} = 4\sqrt{16} = 16$, $\frac{d^2y}{dx^2} = 2 \cdot \frac{2y \frac{dy}{dx} + 14x}{\sqrt{y^2 + 7x^2}} \Big|_{(1,3)} = 2 \cdot \frac{2(3)(16) + 14}{4} = 55$.

3-6-6

35. Let f be the function given by $f(x) = 2x^3 - 5x^2 + 8x + 1$ on the closed interval $[0, 3]$. What is the instantaneous rate of change of f' at $x = 1$?
- (A) 2 ✓
- (B) 4
- (C) 6
- (D) 8

Answer A

Correct. The instantaneous rate of change of the first derivative is the second derivative.

$$f'(x) = 6x^2 - 10x + 8$$

$$f''(x) = \frac{d}{dx}(f'(x)) = 12x - 10$$

$$f''(1) = 12 - 10 = 2$$

36. If $y = e^{x^3}$, then $\frac{d^2y}{dx^2} =$
- (A) $18x^3e^{x^3}$
- (B) $9x^4e^{2x^3}$
- (C) $(6x + 3x^2)e^{x^3}$
- (D) $(6x + 9x^4)e^{x^3}$ ✓

Answer D

Correct. The notation $\frac{d^2y}{dx^2}$ indicates the second derivative of y . Computing the first and second derivatives requires the use of both the chain rule and the product rule.

$$\frac{dy}{dx} = 3x^2e^{x^3}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(3x^2e^{x^3}) = 6x \cdot e^{x^3} + 3x^2 \cdot (3x^2e^{x^3}) = (6x + 9x^4)e^{x^3}$$

3-6-6

37. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant.
- (a) Find $f'(x)$ and $f''(x)$.
- (b) For what value of the constant k does f have a critical point at $x=1$? For this value of k , determine whether f has a relative minimum, relative maximum, or neither at $x = 1$. Justify your answer.
- (c) For a certain value of the constant k , the graph of f has a point of inflection on the x -axis. Find this value of k .

Part A

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for: $f'(x)$

1 point is earned for: $f''(x)$

$$f'(x) = \frac{k}{2\sqrt{x}} - \frac{1}{x}$$

$$f''(x) = -\frac{1}{4}kx^{-3/2} + x^{-2}$$

Part B

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

$$\text{When } k = 2, f'(1) = 0 \text{ and } f''(1) = -\frac{1}{2} + 1 > 0.$$

f has a relative minimum value at $x = 1$ by the Second Derivative Test.



0	1	2	3	4
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The student response earns all of the following points:

3-6-6

1 point is earned if sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned if solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

When $k = 2$, $f'(1) = 0$ and $f''(1) = -\frac{1}{2} + 1 > 0$.

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

Part C

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $f''(x) = 0$ or $f(x) = 0$

1 point is earned for the equation in one variable

1 point is earned for the answer

At this inflection point, $f''(x) = 0$ and $f(x) = 0$.

3-6-6

$$f''(x) = 0 \Rightarrow \frac{-k}{4x^{3/2}} + \frac{1}{x^2} = 0 \Rightarrow k = \frac{4}{\sqrt{x}}$$

$$f(x) = 0 \Rightarrow k\sqrt{x} - \ln x = 0 \Rightarrow k = \frac{\ln x}{\sqrt{x}}$$

$$\text{Therefore, } \frac{4}{\sqrt{x}} = \frac{\ln x}{\sqrt{x}}$$

$$\Rightarrow 4 = \ln x$$

$$\Rightarrow x = e^4$$

$$\Rightarrow k = \frac{4}{e^2}$$